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EXPERIMENT 6

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(A)

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

Sales

sale_date	fruit	sold_num
2020-05-01	apples	10
2020-05-01	oranges	8
2020-05-02	apples	15
2020-05-02	oranges	15
2020-05-03	apples	20
2020-05-03	oranges	0
2020-05-04	apples	15
2020-05-04	oranges	16

Output:

sale_date	diff
2020-05-01	2
2020-05-02	0
2020-05-03	20

2020-05-04	-1

Solution:

```
SELECT
SALE_DATE,
SUM(
   CASE WHEN FRUIT = 'apples' THEN SOLD_NUM
   ELSE -SOLD_NUM
   END
) AS DIFF
FROM
SALES AS S
GROUP BY SALE_DATE
ORDER BY SALE_DATE;
```

(B)

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- id represents the current node.
- p_id represents the parent node.

Classify every node into one of the following categories:

- **Root**: A node that does not have any parent.
- **Leaf**: A node that has a parent but does not have any children.
- **Inner**: A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

Tree

id	p_id
1	NULL
2	1
3	1
4	2
5	2

Output:

id	Type
1	Root

2	Inner
3	Leaf
4	Leaf
5	Leaf

Solution:

```
SELECT
ID,
(
        CASE WHEN P_ID IS NULL THEN 'ROOT'
        WHEN ID NOT IN (
                SELECT P_ID FROM TREE
                        WHERE P_ID IS NOT NULL
        ) THEN 'LEAF'
        ELSE 'INNER'
        END
) AS TYPE
FROM TREE
ORDER BY ID;
```